Jisung Hwang

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Ph.D. Student • KAIST Visual AI Group • School of Computing

EDUCATION

Mar. 2026 ~ Present — **KAIST Visual AI Lab** *Candidate for Ph.D.* — *Daejeon, Republic of Korea*
Supervisor: Minhyuk Sung • KAIST School of Computing

Mar. 2024 ~ Feb. 2026 — **KAIST Visual AI Lab** *M.S.* — *Daejeon, Republic of Korea*
Supervisor: Minhyuk Sung • KAIST School of Computing — GPA : 3.94 / 4.3

Feb. 2020 ~ Feb. 2024 — **KAIST** *B.S.* — *Daejeon, Republic of Korea*
Major: Computer Science (Double: Mathematical Science) — GPA : 3.87 / 4.3
Honors: KAIST Scholarship; National Science & Technology Scholarship due to Academic Excellence

Mar. 2017 ~ Feb. 2020 — **Gyeongbuk Science High School** — *Pohang, Republic of Korea*
Honors: Gyeongsangbuk-do Special Scholarship; IT Education Support Conference Scholarship

AWARDS

2026 — **Outstanding Paper Award — LoViF Workshop @ ECCV 2026**
By the paper 'NoiseTilt: Noise-Tilted Reverse Kernels for Diffusion Reward Alignment'

2026 — **Best Master Thesis Award – KAIST CS Department**
Master Thesis: 'White Gaussian Noise Constraints'

2025 — **Best Master's Paper Award – KCGS 2025**
By the paper 'Occupancy-Based Dual Contouring'

2021 — **Korean National Undergraduate Math Competition**
Bronze Award

2018, 2019 — **Korean Olympiad in Informatics**
Software Sector: Silver Award (2018); Algorithm Sector: Bronze Award (2019)

PUBLICATIONS

2026 — **NoiseTilt: Noise-Tilted Reverse Kernels for Diffusion Reward Alignment**
Jisung Hwang, Yunhong Min, Jaihoon Kim, I-Chao Shen, and Minhyuk Sung — ECCV 2026 — LoViF @ ECCV 2026
Outstanding Paper Award — LoViF Workshop

2026 — **Gradient Preconditioning for Efficient and Reliable Reward-Guided Generation**
Jisung Hwang and Minhyuk Sung — ICML 2026

2026 — **PairFlow: Closed-Form Source-Target Coupling for Few-Step Generation in Discrete Flow Models**
Mingue Park*, Jisung Hwang*, Seungwoo Yoo*, Kyeongmin Yeo, and Minhyuk Sung (*equal contribution) — ICLR 2026

2025 — **Moment- and Power-Spectrum-Based Gaussianity Regularization for Text-to-Image Models**
Jisung Hwang, Jaihoon Kim, and Minhyuk Sung — NeurIPS 2025

2025 — **Inference-Time Scaling for Flow Models via Stochastic Generation and Rollover Budget Forcing**
Jaihoon Kim*, Taehoon Yoon*, Jisung Hwang*, and Minhyuk Sung (*equal contribution) — NeurIPS 2025

2025 — **Neural Green's Functions for Solving Poisson's Equation**
Seungwoo Yoo, Kyeongmin Yeo, Jisung Hwang, and Minhyuk Sung — NeurIPS 2025

2024 — **Occupancy-Based Dual Contouring**
Jisung Hwang and Minhyuk Sung — SIGGRAPH Asia 2024

REVIEWER SERVICE

2026 — **ICML 2026**
2026 — **SIGGRAPH 2026**
2025 — **ICLR 2026**
2025 — **SIGGRAPH Asia 2025**

TEACHING EXPERIENCE

Spring 2026	**CS479 Machine Learning for 3D Data**	*Teaching Assistant, KAIST*
Spring 2025	**CS479 Machine Learning for 3D Data**	*Teaching Assistant, KAIST*
Fall 2024	**CS300 Introduction to Algorithms**	*Teaching Assistant, KAIST*

WORK & LEADERSHIP EXPERIENCE

Dec. 2025 ~ Mar. 2026 — **Visiting at Igarashi Lab, University of Tokyo** — *Tokyo, Japan*
Internship Trainee

- Researched effective inference-time scaling in multi-step generation by tilting the noise

Dec. 2021 ~ May 2023 — **SpiderCore** — *Daejeon, Republic of Korea*
AI Development Team Intern

- Conducted prediction of cell-specific protein–protein interactions by multitask learning
  - Improved the ROC-AUC value by 4–18% on HeLa, HEK293, HCT116, etc.
- Researched cross-dimensional expandability of GNN using multi-dimensional snake game
  - Showed that an RL agent trained in 2D can generalize to higher dimensions
  - Visualized 3D snake game graphically using OpenGL in C++
- Developed and optimized Python and C++ programs to train and evaluate AI models

Jul. 2022 ~ Sep. 2022 — **KAIST-POSTECH Science War**
School Representative for AI Match

- Implemented PPO, AlphaGo, and heuristic algorithms for the modified Quoridor game
- **Led the victory in AI for the first time in 7 years**

Sep. 2022 ~ Sep. 2023 — **Vlab**
AI Team Developer, AI Development Club

EXTRACURRICULAR EXPERIENCE

Sep. 2020 ~ Jul. 2023 — **FingS** ***Fingerstyle Guitar Club***
Practice and Concert Manager, Group Leader

- Trained 10 people every week for 1 year and managed 3 concerts as a sound engineer

Sep. 2022 ~ Present — **Praiser** ***Christian Band Club***
Electric Guitar, Bass Guitar Player

ADDITIONAL INFORMATION

Language	Fluent in Korean; Intermediate in English; Japanese (JLPT N1, July 2026)
Computer	Highly-skilled in Python, C/C++, PyTorch, CUDA, Flask, Linux, Cakewalk Intermediate in Java, C#, Unity, MATLAB, Scala, MySQL, OpenGL
Interests	Music Production, Playing instruments